

API 離心式泵分類

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API 離心式泵分類

1. 泵型式分類代號
2. 說明圖例
3. 泵主零件材料分類

API 離心式泵分類

泵型式分類代號 (Pump Classification Type Identification)

Pump type ^a			Orientation		Type code
Centrifugal pumps	Overhung	Flexibly coupled	Horizontal	Foot-mounted	OH1
				Centreline-supported	OH2
		Rigidly coupled	Vertical in-line with bearing bracket	—	OH3
			Vertical in-line	—	OH4
		Close-coupled	Vertical in-line	—	OH5
			High-speed integrally geared	—	OH6
	Between-bearings	1- and 2-stage	Axially split	—	BB1
			Radially split	—	BB2
		Multistage	Axially split	—	BB3
			Radially split	Single casing	BB4
	Vertically suspended	Single casing	Discharge through column	Diffuser	VS1
				Volute	VS2
				Axial flow	VS3
		Single casing	Separate discharge	Line shaft	VS4
				Cantilever	VS5
		Double casing	Diffuser	—	VS6
			Volute	—	VS7

2014年



API 離心式泵分類

說明圖例

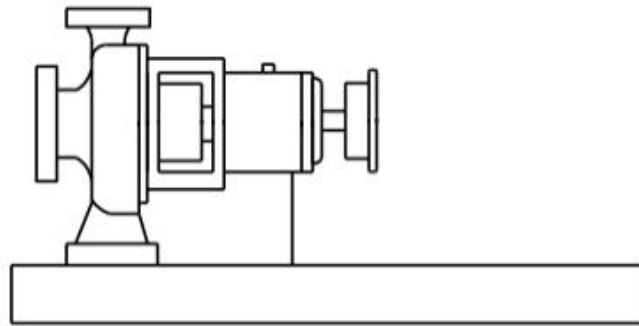


Figure 1 — Pump type OH1

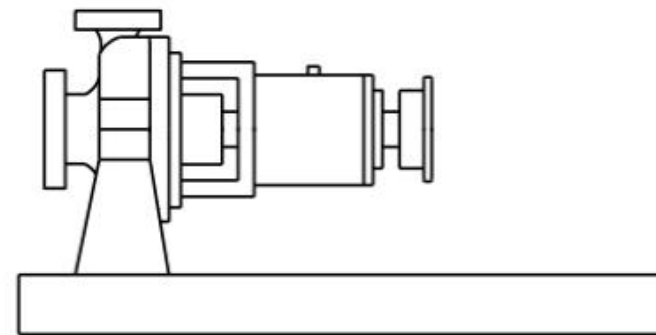


Figure 2 — Pump type OH2

API 離心式泵分類

說明圖例

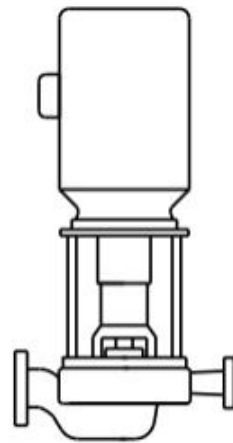


Figure 3 — Pump type OH3

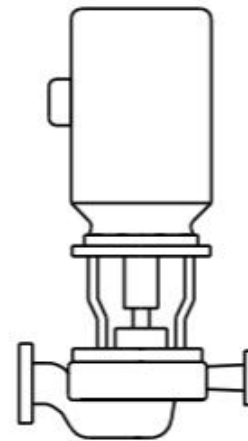


Figure 4 — Pump type OH4

API 離心式泵分類

說明圖例

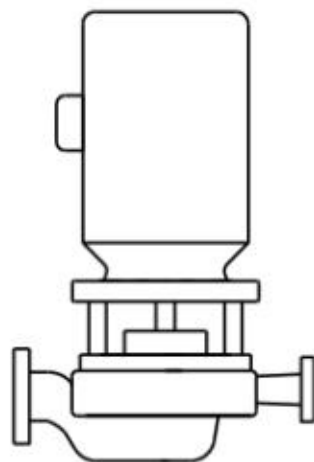


Figure 5 — Pump type OH5

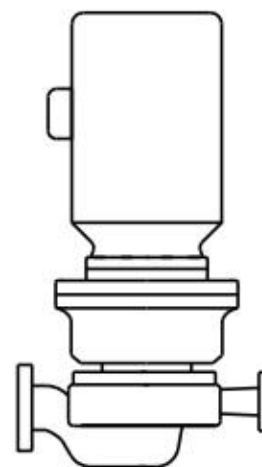


Figure 6 — Pump type OH6

API 離心式泵分類

說明圖例

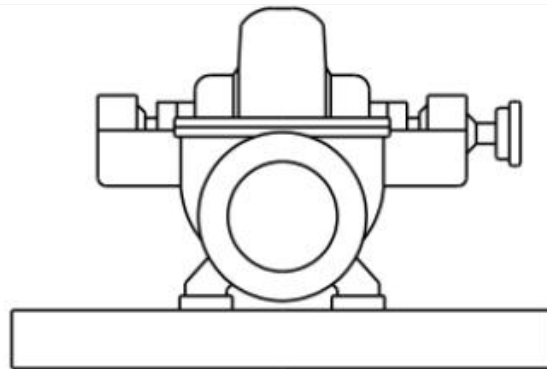


Figure 7 — Pump type BB1

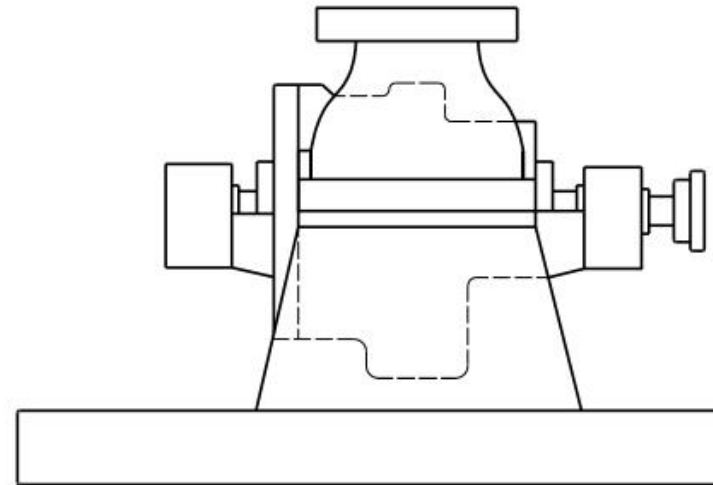


Figure 8 — Pump type BB2

API 離心式泵分類

說明圖例

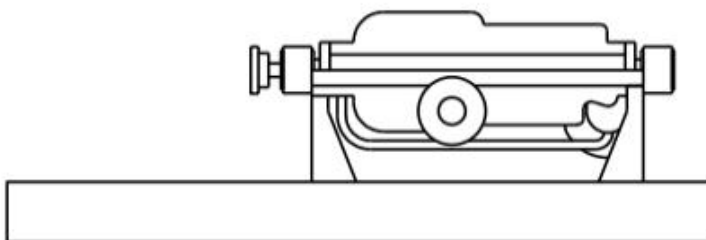


Figure 9 — Pump type BB3

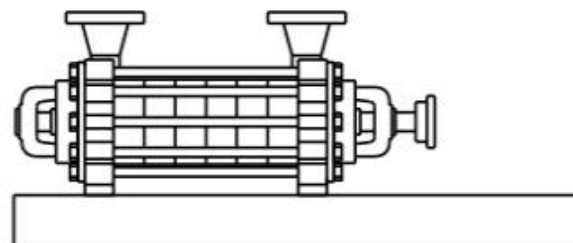


Figure 10 — Pump type BB4

API 離心式泵分類

說明圖例

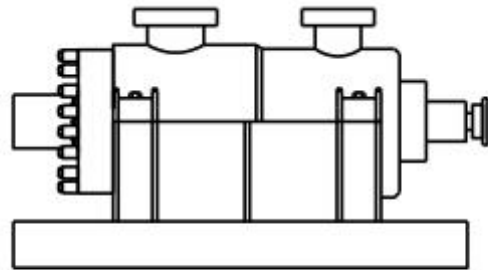


Figure 11 — Pump type BB5

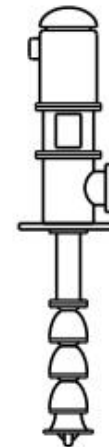


Figure 12 — Pump type VS1

API 離心式泵分類

說明圖例

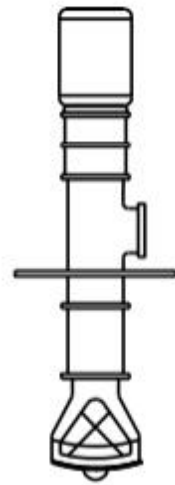


Figure 13 — Pump type VS2



Figure 14 — Pump type VS3

API 離心式泵分類

說明圖例



Figure 15 — Pump type VS4

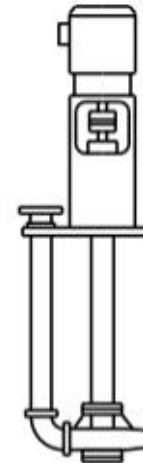


Figure 16 — Pump type VS5

API 離心式泵分類

說明圖例



Figure 17 — Pump type VS6

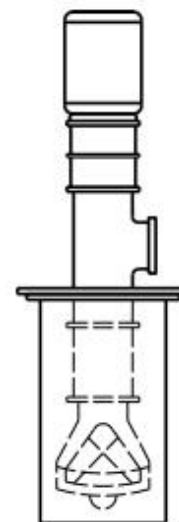


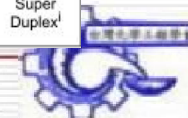
Figure 18 — Pump type VS7

API 離心式泵分類

泵主零件材料分類

Part	Material classes and abbreviations														
	Material class	I-1	I-2	S-1	S-3	S-4	S-5	S-6	S-8 ^l	S-9 ^l	C-6	A-7	A-8	D-1 ^l	D-2 ^l
	Full compliance material ^a	CI ^b	CI	STL	STL	STL	STL	STL	STL	STL	12 % CR	AUS	316 AUS	Duplex	Super Duplex
	Trim material	CI	BRZ	CI	Ni-Resist	STL	STL 12 % CR	12 % CR	316 AUS	Ni-Cu alloy	12 % CR	AUS ^{cd}	316 AUS ^d	Duplex	Super Duplex
Pressure casing	Yes	Cast iron	Cast iron	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	12 % CR	AUS	316 AUS	Duplex	Super Duplex
Inner case parts (bowls, diffusers, diaphragms)	No	Cast iron	Bronze	Cast iron	Ni-Resist	Cast iron	Carbon Steel	12 % CR	316 AUS	Ni-Cu alloy	12 % CR	AUS	316 AUS	Duplex	Super Duplex
Impeller	Yes	Cast iron	Bronze	Cast iron	Ni-Resist	Carbon steel	Carbon Steel	12 % CR	316 AUS	Ni-Cu alloy	12 % CR	AUS	316 AUS	Duplex	Super Duplex
Case wear rings ^k	No	Cast iron	Bronze	Cast iron	Ni-Resist	Cast iron	12 % CR Hardened	12 % CR Hardened	Hard-faced 316 AUS ^e	Ni-Cu alloy	12 % CR Hardened	Hard-faced AUS ^e	Hard-faced 316 AUS ^e	Hard-faced Duplex ^e	Hard-faced Super Duplex ^e
Impeller wear rings ^k	No	Cast iron	Bronze	Cast iron	Ni-Resist	Cast iron	12 % CR Hardened	12 % CR Hardened	Hard-faced 316 AUS ^e	Ni-Cu alloy	12 % CR Hardened	Hard-faced AUS ^e	Hard-faced 316 AUS ^e	Hard-faced Duplex ^e	Hard-faced Super Duplex ^e
Shaft ^d	Yes	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	4140 alloy steel	4140 alloy steel ^f	316 AUS	Ni-Cu alloy	12 % CR	AUS	316 AUS	Duplex	Super Duplex
Throat bushings ^k	No	Cast iron	Bronze	Cast iron	Ni-Resist	Cast iron	12 % CR Hardened	12 % CR Hardened	316 AUS	Ni-Cu alloy	12 % CR Hardened	AUS	316 AUS	Duplex	Super Duplex
Interstage sleeves ^k	No	Cast iron	Bronze	Cast iron	Ni-Resist	Cast iron	12 % CR Hardened	12 % CR Hardened	Hard-faced 316 AUS ^e	Ni-Cu alloy	12 % CR Hardened	Hard-faced AUS ^e	Hard-faced 316 AUS ^e	Hard-faced Duplex ^e	Hard-faced Super Duplex ^e
Interstage bushings ^k	No	Cast iron	Bronze	Cast iron	Ni-Resist	Cast iron	12 % CR Hardened	12 % CR Hardened	Hard-faced 316 AUS ^e	Ni-Cu alloy	12 % CR Hardened	Hard-faced AUS ^e	Hard-faced 316 AUS ^e	Hard-faced Duplex ^e	Hard-faced Super Duplex ^e
Case and gland studs	Yes	4140 alloy steel	4140 alloy steel	4140 alloy steel	4140 alloy steel	4140 alloy steel	4140 alloy steel	4140 alloy steel	4140 alloy steel	Ni-Cu alloy Hardened ^l	4140 alloy steel	4140 alloy steel	4140 alloy steel	Duplex ^l	Super Duplex ^l

2014年



API 離心式泵分類

泵主零件材料分類

Part	Material classes and abbreviations														
	Material class	I-1	I-2	S-1	S-3	S-4	S-5	S-6	S-8 ^l	S-9 ^l	C-6	A-7	A-8	D-1 ^j	D-2 ^j
	Full compliance material ^a	CI ^b	CI	STL	STL	STL	STL	STL	STL	STL	12 % CR	AUS	316 AUS	Duplex	Super Duplex
	Trim material	CI	BRZ	CI	Ni-resist	STL	STL 12 % CR	12 % CR	316 AUS	Ni-Cu alloy	12 % CR	AUS ^{cd}	316 AUS ^d	Duplex	Super Duplex
Case gasket	No	AUS, spiral-wound ^g	AUS, spiral-wound ^g	AUS, spiral-wound ^g	AUS, spiral-wound ^g	AUS, spiral-wound ^g	AUS, spiral-wound ^g	AUS, spiral-wound ^g	316 AUS spiral-wound ^g	Ni-Cu alloy, spiral-wound, PTFE filled ^g	AUS, spiral-wound ^g	AUS, spiral-wound ^g	316 AUS spiral-wound ^g	Duplex SS spiral-wound ^g	Duplex SS spiral-wound ^g
Discharge head/suction can	Yes	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	AUS	AUS	316 AUS	Duplex	Super Duplex
Column/bowl shaft bushings	No	Nitrile butadiene ^h	Bronze	Filled carbon	Nitrile butadiene ^h	Filled carbon	Filled carbon	Filled carbon	Filled carbon	Filled carbon	Filled carbon	Filled carbon	Filled carbon	Filled carbon	Filled carbon
Wetted fasteners (bolts)	Yes	Carbon steel	Carbon steel	Carbon steel	Carbon steel	Carbon steel	316 AUS ^m	316 AUS ^m	316 AUS	Ni-Cu alloy	316 AUS ^m	316 AUS	316 AUS	Duplex	Super Duplex

^a See 6.12.1.4.

^b The abbreviations in the upper part of the second row indicate the case material; the abbreviations in the lower part of the second row indicate trim material. Abbreviations are as follows:

BRZ bronze AUS austenitic stainless steel 316 AUS austenitic stainless steel containing at least 2.0 % molybdenum

STL steel CI cast iron 4140 alloy steel high-strength steels with approximately 0.40 % carbon, 1 % chromium and 0.2 % molybdenum.

12 % CR 12 % chromium

^c Austenitic stainless steels include ISO types 683-13-10/19 (AISI standard types 302, 303, 304, 316, 321, and 347).

^d For vertically suspended pumps with shafts exposed to liquid and running in bushings, the standard shaft material is 12 % chrome, except for classes S-9, A-7, A-8 and D-1. The standard shaft material for cantilever pumps (Type VS5) is 4140 alloy steel where the service liquid allows (see Table G.1).

^e Unless otherwise specified, the requirement for hard-facing and the specific hard-facing material for each application is determined by the vendor and described in the proposal. Alternatives to hard-facing may include opening running clearances (6.7.4) or the use of non-galling materials or non-metallic materials, depending on the corrosiveness of the pumped liquid.

^f For class S-6, the standard shaft material for boiler feed service and for liquid temperatures above 175 °C (350 °F) is 12 % chrome (see Table G.1).

^g If pumps with axially split casings are furnished, a sheet gasket suitable for the service is acceptable. Spiral-wound gaskets should contain a filler material suitable for the service. Gaskets other than spiral-wound may be proposed and furnished if proven suitable for service and approved by the purchaser. See 6.3.10.

^h Alternative materials may be substituted for liquid temperatures greater than 45 °C (110 °F) or for other special services.

ⁱ Unless otherwise specified, AISI 4140 alloy steel may be used for non-wetted case and gland studs.

^j Some applications may require alloy grades higher than the Duplex materials given in Table H.2. "Super Duplex" material grades with pitting resistance equivalency (PRE) values greater than 40 can be necessary.

PRE ≥ 40 , where the PRE is based on actual chemical analysis.

PRE = $w_{Cr} + 3.3w_{Mo} + 16w_{Ni}$, where w is the percentage mass fraction of the element indicated by the subscript.

Note that alternative materials such as "super austenitic" may also be considered.

^k Non-metallic wear part materials, proven compatible with the specified process liquid, may be proposed within the applicable limits shown in Table H.3. Also see 6.7.4 c).

^l The vendor shall consider the effects of differential material expansion between casing and rotor and confirm suitability if operating temperatures can exceed 95 °C (200 °F).

^m For applications where large differences of thermal expansion can result if austenitic stainless steel fasteners are used, alternative fastener materials, such as 12 % or 17 % Cr martensitic steel, with appropriate corrosion resistance, may be used.

2014年



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